

Symmetric Pattern

Custom Feature for Onshape - linear and grid patterns with symmetric margins.

What It Does

Symmetric Pattern creates copies of a selected body or face operation between boundary vertices. Grid mode supports staggered rows, shortening shifted rows from the right side, and manually skipping selected row/column instances.

Inputs

Input	Purpose
Pattern type	Choose Body for whole parts or Faces for selected operation faces such as holes.
Start / End point	Boundary vertices for the column direction.
Columns	Total number of columns, including the original feature.
Mode / Rows	Linear for one row, Grid for multiple rows.
Row end point	Boundary vertex for the row direction. Use a corner or vertex on the intended grid boundary.
Stagger	Optional horizontal offset for every second row: None, 50%, or Custom percentage.
Reduce staggered rows	When Stagger is active, shorten shifted rows by Reduce by. Elements are removed from the right end only.
Skip instances	Manually skip up to 10 instances by row and column. Numbering starts at 1.

Basic Workflow

- 1 Create the original feature, for example one cut or hole.
- 2 Add Symmetric Pattern from the custom feature toolbar and select target geometry.
- 3 Select Start point and End point, then set Columns.
- 4 For Grid mode, set Rows and select Row end point on the opposite row boundary.
- 5 Use Stagger when alternating rows should be shifted horizontally.
- 6 If shifted rows enter the right margin, enable Reduce staggered rows and set Reduce by.
- 7 To remove specific copies, enable Skip instances, set Skip count, then enter row and column pairs such as row 2 column 3.

Grid Placement Rule

Columns and rows use the same margin rule. Row end point is a boundary point, not the center of the last copy. Stagger reduction removes elements only from the right end of shifted rows, so the first shifted element keeps its stagger offset. Skip instances is applied by user-facing row and column numbers after row reductions are considered.

Recommended Test Before Publishing

Test Grid mode with 2, 3, 4, 5, and 7 rows. Test Columns 5, Rows 4, Stagger 50%, Reduce by 1. Then enable Skip instances with row 2 column 3 and row 4 column 5 and confirm only those copies are omitted.

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